CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A fuse is connected in series with the appliance. Reason (R): A fuse must carry the whole current of the appliance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Seeds in cups: A (dry cotton), B (wet cotton, kept in light), C (wet cotton, kept in dark). Which pair isolates the effect of light? 1

(a) B and C **(b)** A and C
(c) A and B **(d)** A, B and C together

3. A student observes an onion peel with 10x eyepiece and 10x objective, and the cell length in the picture is 5 mm. The real length is: 1

(a) 0.005 mm **(b)** 0.05 mm **(c)** 5 mm **(d)** 0.5 mm

4. A town has 20,000 people. If 15,000 are vaccinated, the percentage vaccinated is: 1

(a) 15% **(b)** 75% **(c)** 50% **(d)** 25%

5. Four cells in a circuit are connected one way and the bulb glows. If one cell is reversed, the bulb: 1

(a) melts **(b)** glows dimmer **(c)** does not change **(d)** glows brighter

6. Which of these shows a good use of a control in a test on a new fertiliser? 1

(a) plants without fertiliser grown in same conditions
(b) different plants different soil
(c) no plants
(d) all plants given fertiliser

7. A dough is kept (i) in a warm place, (ii) in a fridge. Dough (i) rises more because: 1

(a) yeast is more active at warm temperature **(b)** the fridge makes yeast grow faster
(c) the cooler dough releases more CO2 **(d)** yeast dies in warmth

8. An outbreak of dengue is reported in a colony. Which of the following steps is least useful? 1

(a) using mosquito repellent **(b)** boiling milk
(c) removing stagnant water **(d)** fogging by authorities

9. An electromagnet lifts a nail when connected to a 3 V cell. With a 6 V battery it lifts more nails because: 1

(a) the current is larger
(b) the coil is longer
(c) the nails are lighter
(d) iron becomes heavier

10. Assertion (A): A hypothesis can be rejected by an experiment. Reason (R): A hypothesis must be a testable statement. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Antibiotics are not given for viral infections such as influenza. Reason (R): Viruses do not have the cell structures targeted by antibiotics. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Malaria can be controlled by draining stagnant water. Reason (R): Mosquito larvae develop in stagnant water. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. **Assertion (A):** A compass needle placed near a wire deflects when current flows through the wire. 1

Reason (R): An electric current produces a magnetic field around the wire.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A hypothesis can be tested by an experiment or observation. 1

Reason (R): A hypothesis is a tentative, testable explanation or prediction.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Plant cells have a definite shape. 1

Reason (R): Plant cells have a rigid cell wall outside the cell membrane.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is Amoeba able to change its shape but a plant cell cannot? 2

17. Why do chronic diseases like diabetes require lifelong care? 2

18. Why do two bulbs in series glow dimly compared with a single bulb? 2

19. Why are tables with units preferred over writing results in sentences? 2

20. Why is the cell called the 'building block of life'? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A bulb is connected through a very long thin wire to a cell. The bulb is dim. Explain and suggest improvement. 3

22. Suggest how you will test the claim that a particular kind of paper towel absorbs more water than another. 3

23. A farmer sprays an antibiotic on a crop disease caused by a fungus and it fails. Why? 3

24. Design a three-point health plan for a school hostel to avoid infection. 3

OR

Why do electrical appliances not work if the switch is turned off, though the plug is still in the socket? 3

25. A student records 5 readings of the time taken for a ball to roll: 4.1, 4.0, 4.2, 6.5, 4.1 s. Identify the outlier, suggest its cause and give the best mean. 3

OR

Differentiate between a virus and a bacterium based on size, structure and treatment. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** In the school laboratory, Kavya peels a thin layer from an onion, places it on a slide, adds a drop of safranin stain and a cover slip, and observes it under a compound microscope. She sees rectangular cells arranged closely, each with a dark round spot near the edge. 5

(i) Why is the peel kept thin?

(ii) Name the dark round structure.

(iii) Name the outer rigid layer that gives the rectangular shape.

(iv) She then scrapes cells from the inside of her cheek. Name two differences she would see compared with onion cells.

27. Explain the concepts: pathogen, host, vector, symptoms, immunity, vaccine, antibiotic, epidemic using one line each. 5

OR

Compare how a cell, a battery and mains supply differ in voltage and risk. Explain why a cell is safe to touch but the mains is not. 5

END OF QUESTION PAPER — Total: 50 Marks